



THE MAHARAJA SAYAJIRAO UNIVERSITY OF BARODA

Faculty of Science | Department of Statistics |

M.Sc. Previous (2025-26)

Public Awareness and Behavioral Intentions for Climate Change Mitigation of Vadodara City

A Statistical Project Report

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DECLARATION:

We hereby declare that the work presented in this project titled "Public Awareness and Behavioral Intentions for Climate Change Mitigation of Vadodara City" is the result of our independent work, submitted to the Department of Statistics, The Maharaja Sayajirao University of Baroda, as a part of our journey towards earning our Master of Science degree. This project was undertaken under the expert guidance of Dr. Manisha Pandya.

This project showcases our original research efforts and represents a significant milestone in our academic journey. The findings presented here have not been shared with any other University or Institute for the award of any degree.

Place: Vadodara

Team Members:

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CERTIFICATE :

This is to certify that Mahak Garg, Jaymeet Modi, Darshan Bokadiya, and Rutvik Chaudhari have successfully and satisfactorily completed the project titled:

"Public Awareness and Behavioral Intentions for Climate Change Mitigation of Vadodara City"

As a team for the academic year 2025–26, they have submitted the work to the Department of Statistics as a partial fulfilment of the requirements for the Master's Degree in Statistics. The work presented is their original work conducted under the supervision and guidance of Dr. Manisha Pandya.

I wish them grand success in the future.

GUIDE BY:

Dr. Manisha Pandya

Head of Department
Prof.(Dr.)Vipul Kalamkar

ACKNOWLEDGMENT

First and foremost, we would like to express our heartfelt gratitude to our guide, Dr. Manisha Pandya, who consistently helped us, welcomed our questions, kept us motivated, and provided invaluable recommendations and suggestions. This project would not have been possible without her unwavering support, advice, and guidance.

We also extend our gratitude to the Head of the Department of Statistics for their support and helpful feedback throughout this academic journey.

We would like to thank The Maharaja Sayajirao University of Baroda for providing us with excellent faculty, library resources, and the academic environment necessary for our growth.

Our sincere thanks to our parents for their constant support and guidance at every stage. Finally, we extend our gratitude to all the 304 respondents of Vadodara who participated in our survey, and to our seniors and friends for their patience and encouragement.

ABSTRACT

This study examines public awareness and behavioral intentions related to climate change mitigation among residents of Vadodara city. With climate change emerging as one of the most pressing global challenges, understanding community-level awareness, attitudes, and behavioural responses is crucial for effective policy formulation.

Primary data was collected from 304 respondents across four geographical zones (North, South, East, and West) of Vadodara using stratified random sampling. The study explores awareness levels across education groups, the relationship between awareness and eco-friendly behaviour, barriers to sustainable practices, and perceived sources of carbon emissions.

Secondary data analysis covers temperature trends in Vadodara (2010–2025), the relationship between population growth and CO₂ emissions (2010–2023), and sector-wise carbon emission contribution. Statistical methods include Chi-Square Test of Independence, Goodman-Kruskal Gamma Test, Pearson's Standardized Residuals, and Spearman Rank Correlation.

Key findings reveal significant association between education level and climate awareness, strong positive correlation between awareness and recycling behaviour, dominance of industry and transportation as emission sources, and a statistically significant upward trend in maximum temperatures.

Keywords: Climate Change, Public Awareness, Carbon Emissions, Vadodara, Eco-Friendly Behaviour, Statistical Analysis, Stratified Sampling.

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INTRODUCTION

Climate change is one of the most critical environmental challenges confronting our planet today. It refers to long-term alterations in global temperature patterns, precipitation levels, and weather events—primarily driven by anthropogenic activities. Rapidly increasing carbon dioxide emissions from industrial processes, deforestation, and the widespread use of fossil fuels have accelerated the pace of climate change to an unprecedented level.

The effects of climate change are far-reaching, impacting agriculture, water resources, public health, biodiversity, and economic systems. In urban areas like Vadodara—a rapidly growing industrial city in Gujarat—these impacts are increasingly visible through rising temperatures, irregular rainfall, and deteriorating air quality.

This project focuses on two dimensions of the climate crisis: the level of public awareness about climate change, and the willingness of residents to adopt climate-mitigating behaviours. By examining how aware people are and what prevents them from taking action, this study aims to provide actionable insights for policymakers, educators, and community leaders.

Two critical analytical dimensions are considered: carbon emissions (both household and sector-level) and the adoption of eco-friendly products and practices. The study integrates primary survey data from 304 Vadodara residents with secondary climate and emission data from 2010 to 2025.

WHY THIS PROJECT?

- Climate change affects every sector — health, economy, agriculture, and daily life. Yet public behavioral response remains insufficient.
- Vadodara is a rapidly growing urban-industrial city experiencing rising temperatures and increasing carbon emissions.
- Studying public awareness helps understand whether citizens are taking climate change seriously and translating knowledge into action.
- This research helps policymakers design more effective awareness campaigns tailored to different education and demographic groups.
- It provides an empirical comparison between public perception of emission sources and actual scientific data on sectoral contributions.
- With limited Indian urban-level studies combining survey and secondary emission data, this work fills an important research gap.

OBJECTIVES OF THE STUDY

PRIMARY OBJECTIVES:

1. To measure the average level of public awareness about climate change among different education groups.
2. To analyze the relationship between awareness level and eco-friendly behaviour (recycling, saving electricity, using public transport).
3. To identify the key barriers preventing residents from adopting sustainable lifestyle practices.
4. To identify major sources of carbon emissions as perceived by the public.

SECONDARY OBJECTIVES:

1. To analyze the trend of average annual temperature change in Vadodara (2010-2025).
2. To study the relationship between population growth and CO₂ emissions in Vadodara.
3. To identify the major contributing sectors to carbon emissions within the Vadodara region.

METHODOLOGY AND DATA

Type of Data: Primary data was collected through a structured self-administered questionnaire distributed across four zones of Vadodara city. The questionnaire comprised 17 questions covering demographics, climate awareness, eco-friendly behaviour, carbon emission perception, and government policy effectiveness. Secondary data was obtained from official meteorological records and published emission inventories for Vadodara (2010–2025).

Sampling Method: A geographical stratification approach was adopted to ensure representative coverage of all parts of Vadodara city. The city was divided into four geographical zones: North, South, East, and West. From each zone, one ward was randomly selected, and respondents were surveyed proportionally based on ward population.

Questionnaire Design: The questionnaire included questions on a 5-point scale (for knowledge rating and concern levels), multiple-choice questions (for barrier identification and emission source perception), and demographic questions. The instrument was pre-tested with a pilot sample of 50 respondents

Tools Used: Data collection was done through both physical (hard copy) and digital (Google Form) methods. Statistical analysis was performed using R Programming (for hypothesis testing), python and Microsoft Excel (for data processing and visualization).

SAMPLE SIZE DETERMINATION

Sample size was determined using the Cochran formula for estimating a population proportion with a specified margin of error:

$$n = Z^2 \alpha / 2 \times p(1-p) / E^2$$

Parameter	Value
p (Pilot proportion)	37/50 = 0.74
Z (Critical value at 95% CI)	1.96
E (Margin of error)	0.05
Calculated sample size (n)	295
Actual responses collected	304

Interpretation: Based on a pilot study of 50 respondents (37 aware of climate change, $p = 0.74$), the required minimum sample size was calculated as 295. A total of 304 valid responses were collected, exceeding the minimum requirement and ensuring adequate statistical power.

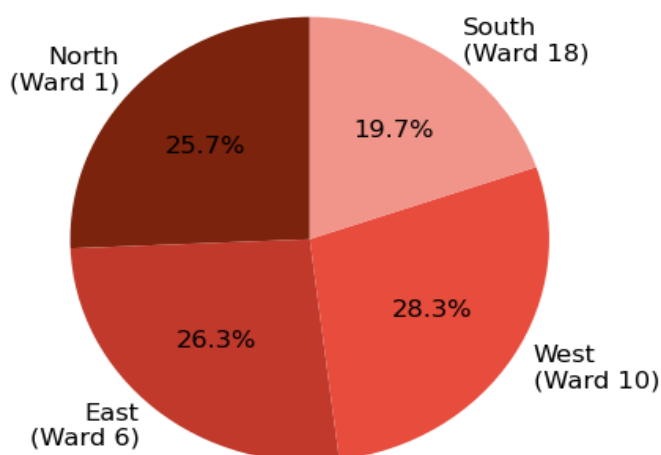
SAMPLING PROCEDURE – ZONE & WARD ALLOCATION

Vadodara city was divided into four geographical zones. One ward was selected from each zone using stratified random sampling. Sample allocation was proportional to ward population.

Zone	Selected Ward	Ward Population	Sample Size
North	Ward 1	4,35,914	78
East	Ward 6	4,56,753	80
West	Ward 10	4,87,117	86
South	Ward 18	3,62,007	60
TOTAL	4 Wards	17,41,791	304

Sample Distribution by Zone

Interpretation: Ward 10 (West zone) contributes the largest sample share (28.3%), reflecting its higher population. Ward 18 (South) has the smallest share (19.7%).



Chi-Square Test of Independence:

The Chi-Square Test of Independence is focused to examine whether two categorical variables are associated or independent of each other.

Project purpose:

Chi-Square Test of Independence (Education vs Awareness)

The Chi-Square Test of Independence was applied to examine the relationship between education level and awareness level of climate change. Both variables are categorical in nature, with awareness level being ordinal but still falling under the category of categorical data.

This test is appropriate because it evaluates whether the observed frequencies differ significantly from the expected frequencies under the assumption of independence. Since the objective was to determine whether education influences awareness, the Chi-Square test provides a valid and robust method to test this association.

Chi-Square goodness of fit :

The Chi-Square Goodness of Fit test is used to check whether the observed data follows a particular expected distribution. The main objective is to test whether all categories occur equally or follow a specified pattern.

Project purpose:

Chi-Square Goodness of Fit Test (Barriers & Emission Sources)

The Chi-Square Goodness of Fit test was applied to examine whether different categories (barriers and emission sources) are equally distributed among respondents.

This test is justified because the objective was to compare observed frequencies with expected equal frequencies. It helps in determining whether the differences observed are statistically significant or due to random variation.

Spearman Rank Correlation :

Spearman Rank Correlation is a non-parametric method used to measure the strength and direction of association between two variables based on their ranks. It does not require normal distribution and is suitable for ordinal data or non-linear relationships.

Project purpose:

Spearman Rank Correlation (Temperature Trend)

Spearman Rank Correlation was used to analyze the trend in temperature data over time. Since time series data may not follow normal distribution and may contain non-linear patterns, a non-parametric test like Spearman is appropriate.

This method is justified because it measures the monotonic relationship between time and temperature without assuming normality.

Pearson Correlation :

Pearson Correlation measures the linear relationship between two continuous variables. It is a parametric test and assumes that the data is normally distributed..

Project purpose:

Pearson Correlation (Population vs CO₂ Emissions)

Pearson Correlation was applied to measure the linear relationship between population growth and CO₂ emissions. Both variables are continuous and quantitative.

This test is justified because it provides a precise measure of linear association and helps in understanding how strongly population growth influences emissions.

Arima Model for forecasting (using python) :

ARIMA Model for Forecasting (Using Python)

The ARIMA (AutoRegressive Integrated Moving Average) model is a widely used statistical technique for analyzing and forecasting time series data. In this project, ARIMA modeling was implemented using Python to predict future carbon emissions for different sectors such as transport, agriculture, industry, and waste for the period 2024–2030.

The ARIMA model works by using past values and past errors of a time series to estimate future values. It is particularly useful when data shows trend or temporal dependency.

Co-occurrence analysis :

Co-occurrence analysis is a statistical technique used to study how frequently two or more variables or categories appear together in a dataset. In simple terms, it shows which factors or responses tend to occur together more often than expected. This method is commonly used in behavioral studies, text analysis, and survey data analysis.

The analysis is usually represented in the form of a co-occurrence matrix, where each cell shows the frequency with which two variables occur together...

Project purpose:

Co-occurrence analysis was used to identify relationships between different eco-friendly behaviours and awareness levels. It helped in understanding which behaviours (such as recycling, saving electricity, and using public transport) tend to occur together among respondents.

Pearson's Standardized Residuals :

Pearson's Standardized Residuals are used after performing a Chi-Square Test of Independence to identify which specific cells contribute most to the overall chi-square value.

Project purpose:

Pearson's Standardized Residuals

After establishing a significant association using the Chi-Square test, Pearson's Standardized Residuals were computed to identify which specific cells contributed most to the overall association.

This method is justified because the Chi-Square test only indicates the presence of association but does not provide detailed insight into the pattern of association. Residual analysis helps in pinpointing the exact categories responsible for the relationship.

<1.96	Positive association
>1.96	Negative association
-1.96 to 1.96	Not significant

GOODMAN – KRUSKAL TEST OF ASSOCIATION:

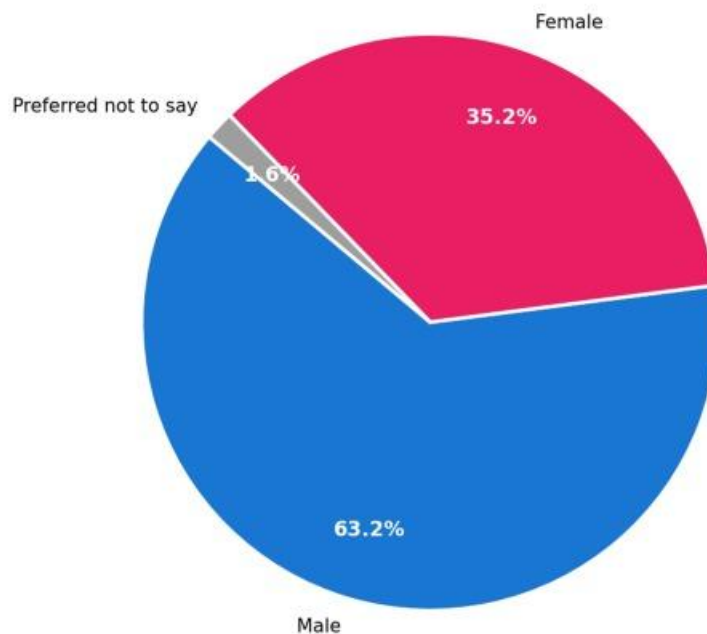
Goodman-Kruskal Gamma Test for Ordinal Categorical Variables is an alternative to the Chi-Square Test of association and is used to test whether there exists a significant association between two ordinal variables or not.

Gamma : The Goodman-Kruskal Gamma value lies between -1 and +1. The nature of the association is interpreted by the sign (negative or positive association) and the strength is given by the absolute value of gamma as follows:

value	Interpret
0.10-0.20	weak
0.20-0.40	moderate
0.40-0.60	relatively strong
0.60-0.80	strong
0.80-1.00	very strong

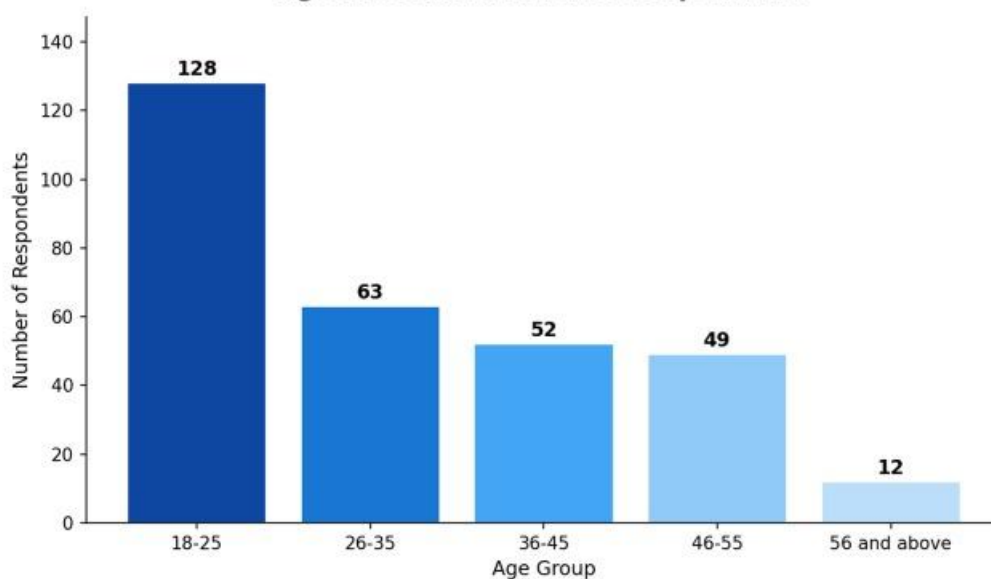
EXPLORATORY DATA ANALYSIS – DEMOGRAPHICS

Gender Distribution of Respondents



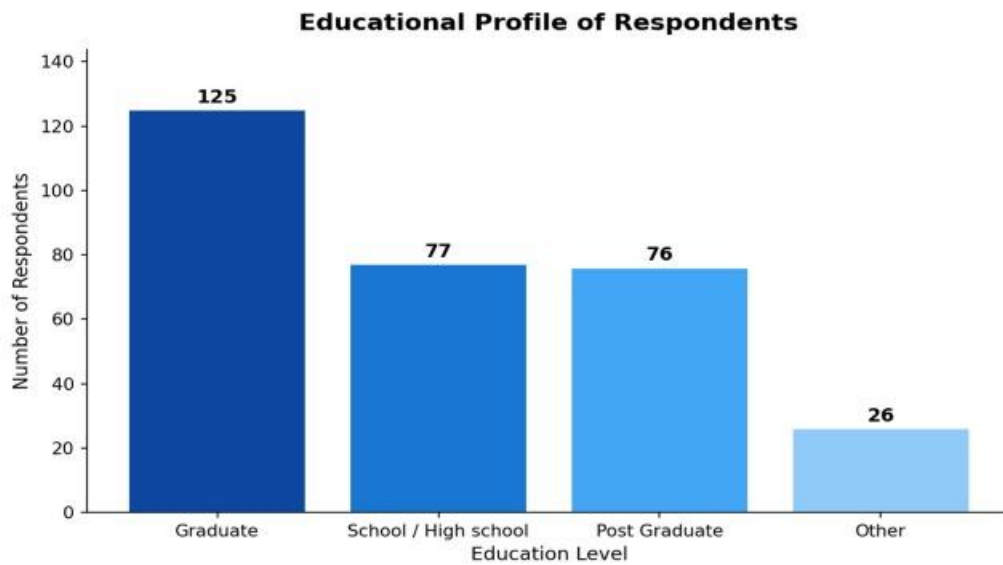
Interpretation: Male respondents are higher in number compared to female respondents, while very few fall in other categories. The sample is male-dominated, so results may be slightly influenced by male perspectives.

Age-wise Distribution of Respondents

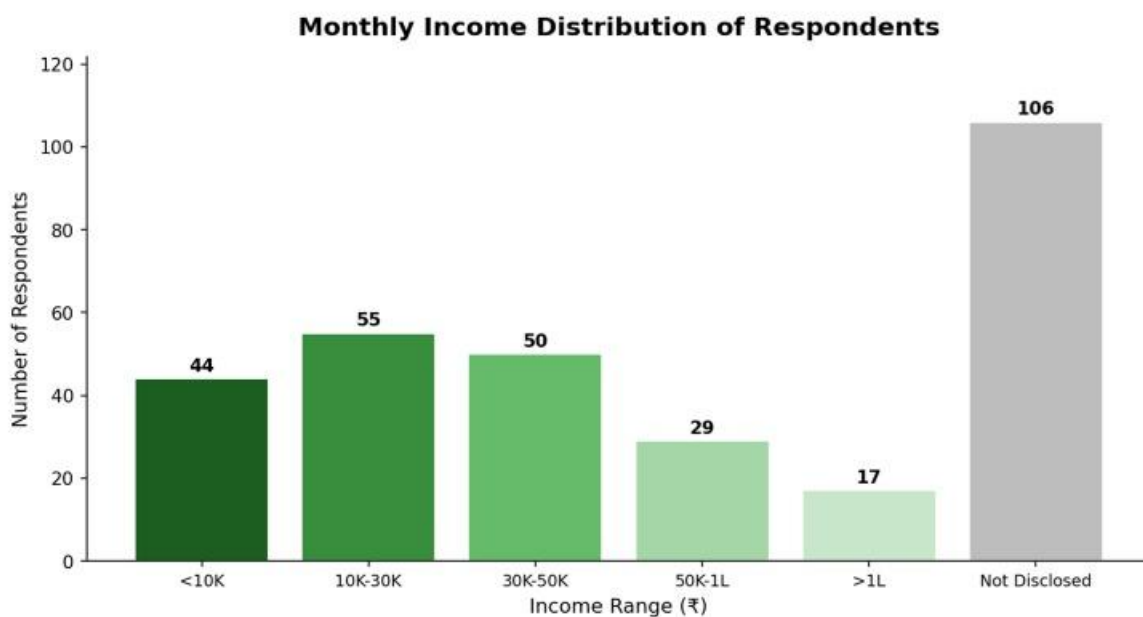


Interpretation: The majority of respondents belong to the 18–25 age group, showing higher participation from young individuals. Older age groups are less represented in the data. The results mainly reflect the views and behaviour of the younger population.

EXPLORATORY DATA ANALYSIS – DEMOGRAPHICS



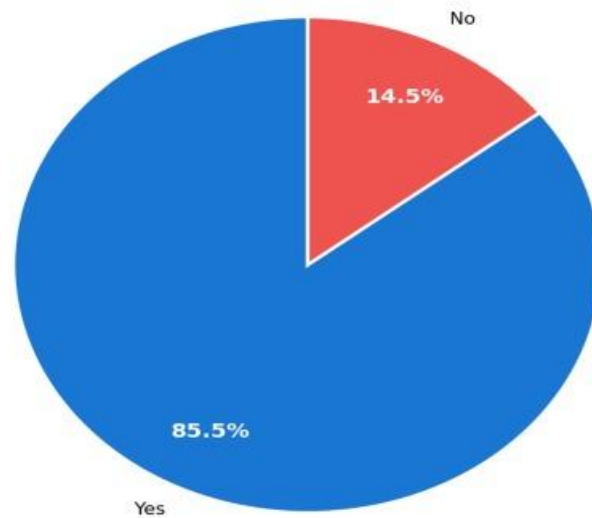
Interpretation: Most respondents are graduates and postgraduates, with fewer from school-level education. This shows a well-educated sample group. Higher education levels may contribute to better understanding of climate issues.



Interpretation: Most respondents fall in low to middle income categories, with fewer high-income individuals. Financial capacity may affect adoption of eco-friendly practices.

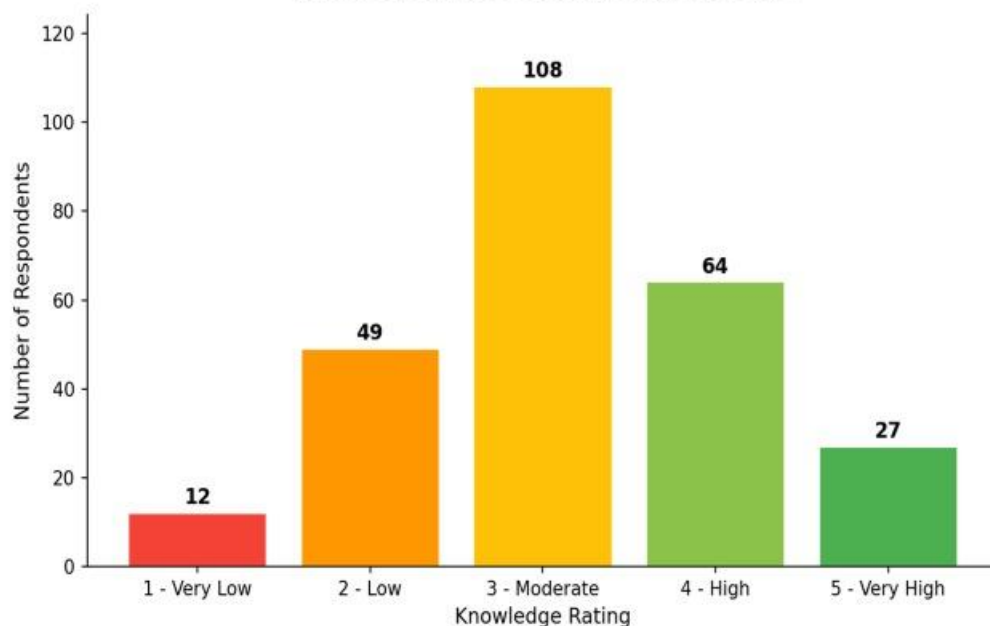
EXPLORATORY DATA ANALYSIS – DEMOGRAPHICS

Awareness about Climate Change Among Respondents



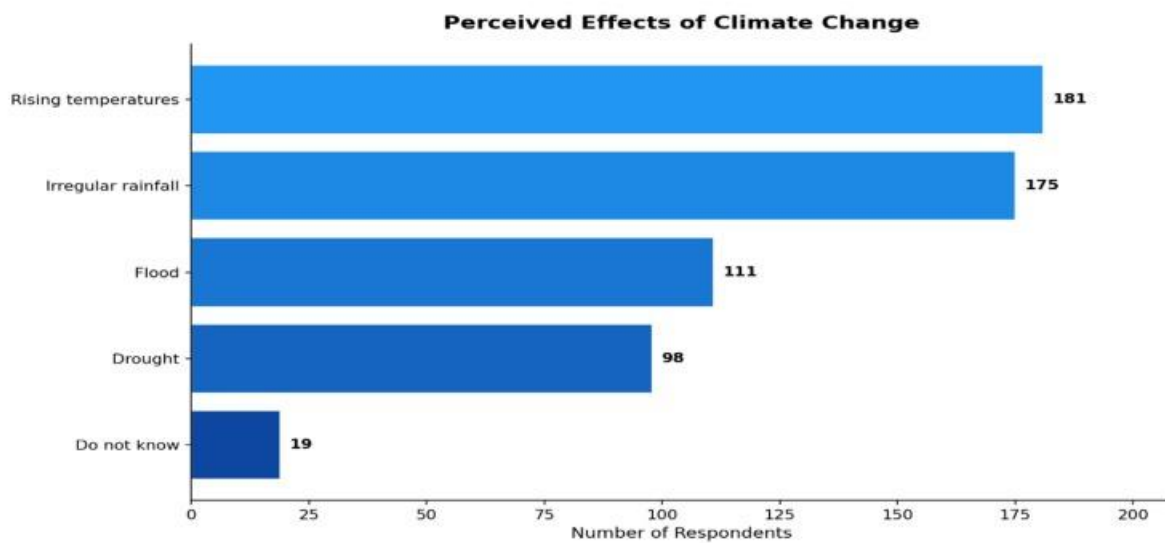
Interpretation: Most respondents answered Yes indicating that they are aware of climate change, while only a small number answered No. This shows that climate change awareness is widespread among the population.

Level of Climate Change Knowledge

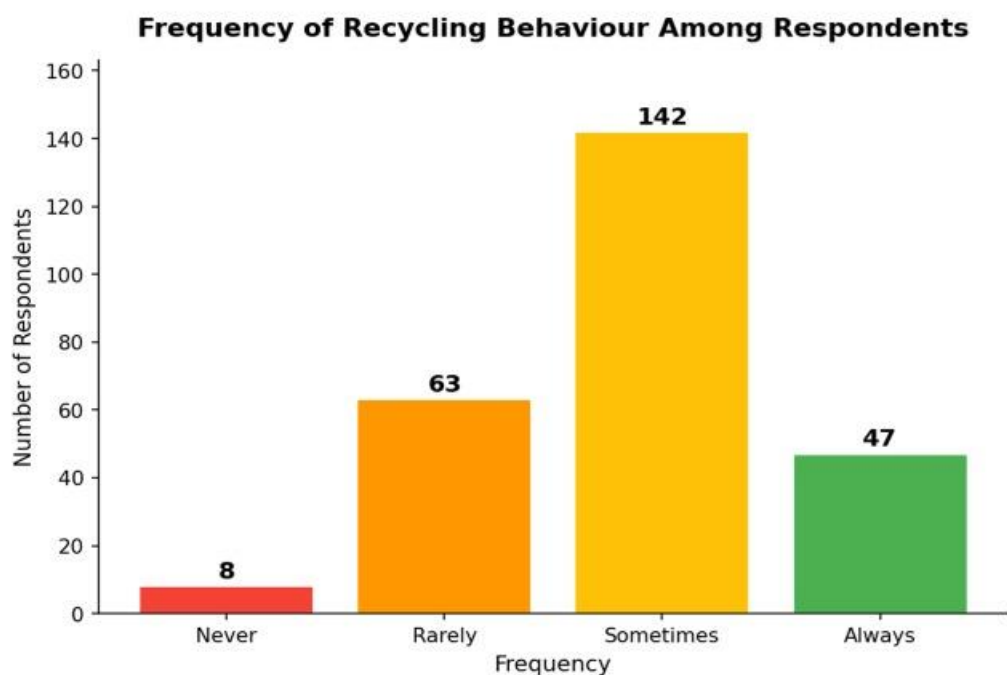


Interpretation: large portion of respondents have moderate to high knowledge about climate change. Very few fall in the low knowledge category. Overall awareness about climate change is fairly good.

EXPLORATORY DATA ANALYSIS – DEMOGRAPHICS

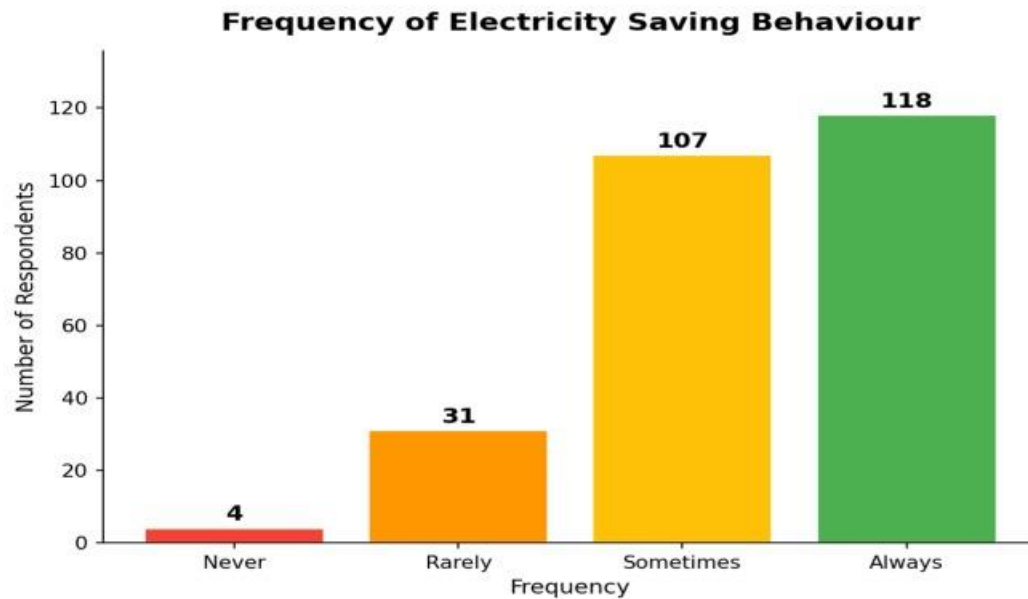


Interpretation: Most respondents identify rising temperature as the most common effect of climate change, followed by issues like floods and irregular rainfall. Fewer respondents mention other impacts.

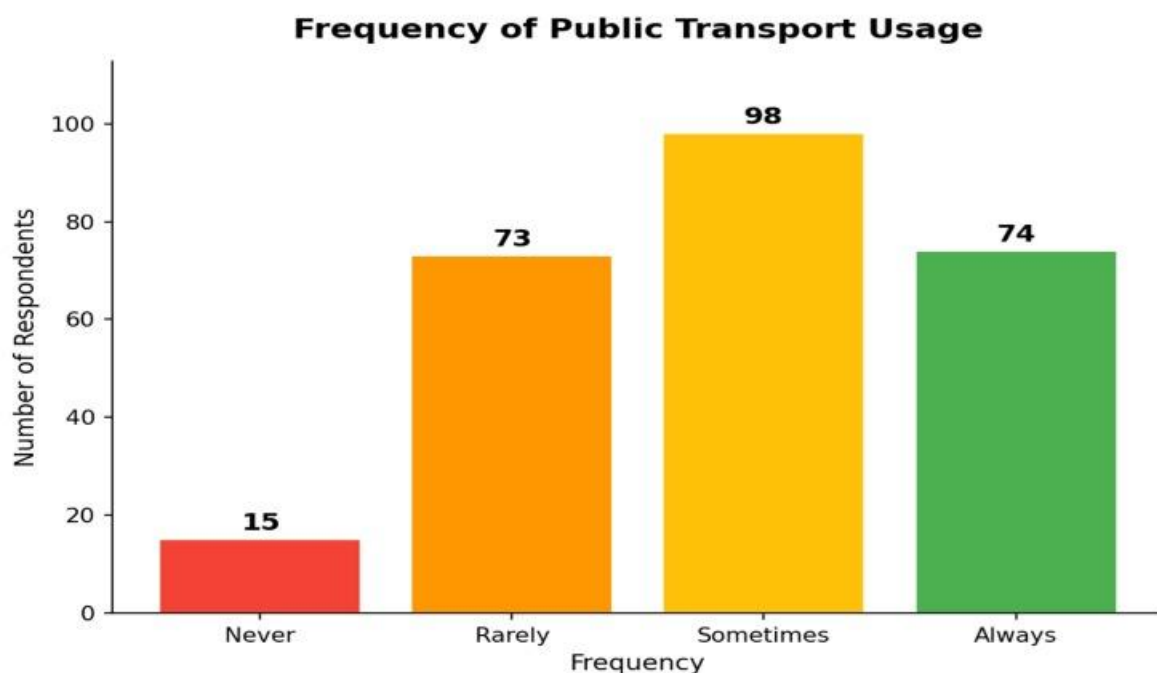


Interpretation: Responses are spread across categories, but many respondents recycle sometimes or often. Fewer respondents never recycle. Recycling habits are present but not consistently practiced by everyone.

EXPLORATORY DATA ANALYSIS – DEMOGRAPHICS

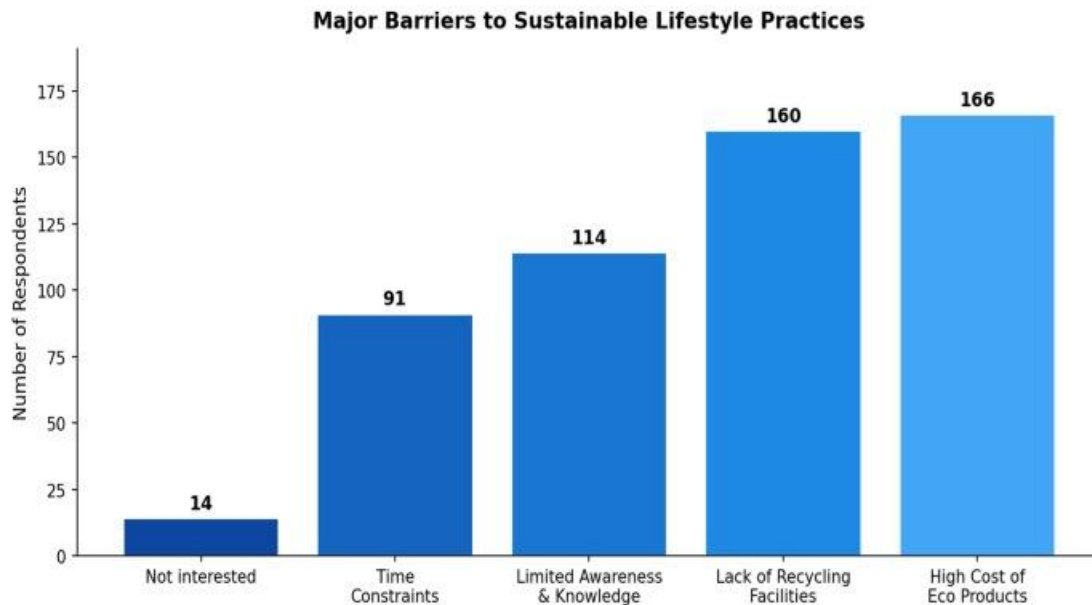


Interpretation: A large number of respondents report saving electricity frequently or always. Energy-saving behaviour is relatively common among respondents

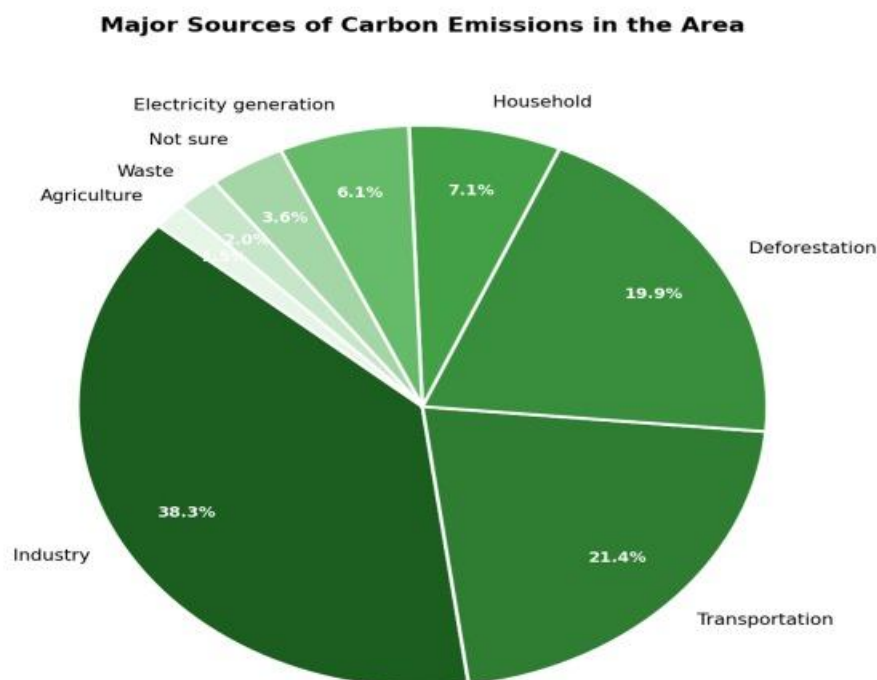


Interpretation: Most respondents use public transport occasionally rather than regularly. Public transport is not the primary choice for daily travel..

EXPLORATORY DATA ANALYSIS – DEMOGRAPHICS



Interpretation: High cost and lack of recycling facilities are reported as the main barriers. Other factors include limited knowledge and time constraints.



Interpretation: Industry is identified as the main source, followed by transportation and deforestation. Some sectors like agriculture and waste are less recognized.

PRIMARY OBJECTIVE 1 – EDUCATION vs. AWARENESS LEVEL

Objective: To measure the average level of public awareness about climate change among different education groups using the Chi-Square Test of Independence.

H₀: Education Level and Awareness Level are independent..

H₁: Education Level and Awareness Level are not independent..

Observed Frequency Table:

Education Level	Low (1-2)	Moderate (3)	High (4-5)	Total
Graduate	22	55	41	118
Post Graduate	13	24	38	75
School / High School	21	26	8	55
Column Total	56	105	87	248

Expected Frequency Table ($E = R_i \times C_j / N$):

Education Level	Low	Moderate	High
Graduate	26.65	49.96	41.40
Post Graduate	16.94	31.75	26.31
School / High School	12.42	23.29	19.29

Test Statistic (χ^2)	Degrees of Freedom	p-value	Decision ($\alpha = 0.05$)
22.1797	4	0.000185 ($\approx 1.85 \times 10^{-4}$)	Reject H ₀

- **Overall conclusion:** since the p value is less than 0.05, the null hypothesis of independence is rejected. Therefore, there is a statistically significant association between education level and awareness level.

PRIMARY OBJECTIVE 1:

Pearson's Standardized Residuals: Association between education level and knowledge level

Cell (i,j)	Observed (N _{ij})	Expected (U _{ij})	Residual
Graduate – Low	22	26.645	-1.413
Graduate – Moderate	55	49.960	+1.297
Graduate – High	41	41.395	-0.105
Post Grad – Low	13	16.935	-1.301
Post Grad – Moderate	24	31.754	-2.170
Post Grad – High	38	26.310	+3.386
School – Low	21	12.419	+3.137
School – Moderate	26	23.286	+0.839
School – High	8	19.294	-3.617

Interpretation:

- Post Graduate respondents are most associated with High Knowledge (residual = +3.386, exceeds +1.96).
- School/High School respondents are most associated with Low Knowledge (+3.137 , exceeds +1.96).
- Graduate respondents show no significant association (all residuals within ± 1.96).

Overall, higher education level is associated with higher climate change awareness, while lower education level corresponds to lower awareness level.

PRIMARY OBJECTIVE 2 – AWARENESS vs. ECO-FRIENDLY BEHAVIOUR

Objective: To analyze the relationship between awareness level and eco-friendly behaviour using the Goodman-Kruskal Gamma Test (for ordinal variables).

H₀: Awareness Level and Recycling Behaviour have no association.

H₁: Awareness Level and Recycling Behaviour have association

Contingency Table – Awareness Level vs. Recycling Behaviour:

Awareness	Never	Rarely	Sometimes	Often	Always	Total
Low	25	21	12	4	1	63
Moderate	5	10	15	9	6	45
High	2	6	15	25	34	82
Total	32	37	42	38	41	190

Result: Gamma (γ) = 0.7348 | p-value = 0.000843

Interpretation: Since p-value (0.000843) < 0.05, we reject H₀ at 5% significance level. The Gamma value of 0.7348 indicates a strong positive association between awareness level and recycling behaviour. Respondents tend to recycle more frequently. This suggests awareness campaigns can significantly improve recycling behaviour.

PRIMARY OBJECTIVE 2

H₀: Awareness Level and saving electricity usage behaviour have no association.

H₁: Awareness Level and saving electricity usage behaviour have association

Contingency Table – Awareness Level vs. saving electricity Behaviour:

Awarene ss	Never	Rarel y	Sometim es	Often	Alwa ys	Total
Low	13	17	16	10	7	63
Moderate	2	5	12	14	12	45
High	1	4	11	25	41	82
Total	16	26	39	49	60	190

Result: Gamma (γ) = 0.5935 | p-value = 0.000032

Interpretation: : Since p-value < 0.05, we reject H₀ at 5% significance level. The Gamma value indicates a relatively strong positive association between awareness level and saving electricity usage behaviour.

Respondents tend to saving electricity more frequently. This suggests awareness campaigns can significantly improve saving electricity behaviour.

PRIMARY OBJECTIVE 2

H₀: Awareness Level and public transport usage behaviour have no association.

H₁: Awareness Level and public transport usage behaviour have association

Contingency Table – Awareness Level vs. public transport Behaviour:

Awareness	Never	Rarely	Sometimes	Often	Always	Total
Low	19	19	13	8	4	63
Moderate	6	9	13	10	7	45
High	6	10	19	22	25	82
Total	31	38	45	40	36	190

Result: Gamma (γ) = 0.4134 | p-value = 0.07

Interpretation: Since p-value > 0.05, we reject H₀ at 5% significance level. This indicates that there is not enough statistical evidence to conclude that a significant relationship exist between awareness level and public transport usage behaviour.

In other words, variations in awareness level do not appear to significantly influence the usage of public transport

PRIMARY OBJECTIVE 3 – Major barriers as a sustainable life practices.

To identify the major barriers as a sustainable life practices by Vadodara residents. Chi-Square Goodness-of-Fit test was applied to check if sectors are equally identified.

Frequency and percentage analysis:

barrier	Frequency
High cost of eco product	166
Lack of recycling	160
Limited knowledge	114
Time constraints	91
Not interested	14

Interpretation : commonly barrier is high cost of eco-products.

Lack of recycling facilities is also major barrier.

Limited awareness acts as an another important barrier.

Very few respondents reported lack of interest as a barrier

Co-occurrence Analysis:

Barrier	High cost of eco product	Lack of recycling facilities	Limited knowledge	Time constraints	Not interested
High cost of eco product	166	103	72	57	3
Lack of recycling facilities	103	160	69	58	3
Limited knowledge	72	69	114	37	2
Time constraint	57	58	37	91	2
Not interested	3	3	2	2	14

INTERPRETATION:

The dominant barriers

High cost of eco products (166) and lack of recycling facilities (160) have the highest diagonal values → these are the most frequently cited individual barriers.

Not interested (14) is very low → most people *do care*, but face practical obstacles.

Strong relationships between barriers

The largest off-diagonal values show which barriers tend to occur together:

High cost ↔ Lack of recycling facilities (103)

People facing high costs also often lack infrastructure. This suggests system-level barriers, not just personal ones.

PRIMARY OBJECTIVE 3

Chi square goodness of fit:

H₀: all barrier are equally important .

H₁: all barrier are not equally important .

Sector	Frequency	Expected frequency
High cost of eco product	166	109
Lack of recycling	160	109
Limited knowledge	114	109
Time constraints	91	109
Not interested	14	109

Test Statistic (χ^2)	p-value	Degrees of Freedom	Decision ($\alpha = 0.05$)
139.66	0.00000000000000003	4	Reject H ₀

Interpretation: since $p\text{-value} < 0.05$, this indicates that all barriers are not equally important.

Frequency analysis identified the barriers with the highest responses ,while chi square goodness of fit test confirmed that the observed differences among barriers were statistically significant and not due to chance.

PRIMARY OBJECTIVE 4 – PERCEIVED SOURCES OF CARBON EMISSIONS

Objective: To identify the major sources of carbon emissions as perceived by Vadodara residents. Chi-Square Goodness-of-Fit test was applied to check if sectors are equally identified.

Frequency and percentage analysis:

Sector	Frequency	Percentage (%)
Industry	75	38.27%
Transportation	42	21.43%
Deforestation	39	19.90%
Household	14	7.14%
Electricity Generation	12	6.12%
Not Sure	7	3.57%
Waste	4	2.04%
Agriculture	3	1.53%
TOTAL	196	100%

Interpretation:

Industry is perceived as major sources of carbon emission with highest percentage

Transport and Deforestation are also considered important contributors to carbon emission.

Agriculture and Waste are perceived as less significant sources of carbon emission.

PRIMARY OBJECTIVE 4 –

Chi square goodness of fit:

H₀: all emission source are equally important

H₁: all emission source are not equally important .

Sector	Frequency	Expected frequency
Industry	75	32.67
Transportation	42	32.67
Deforestation	42	32.67
Household	14	32.67
Electricity Generation	12	32.67
Waste	11	32.67

Test	χ^2 Value	df	p-value	Decision
Chi-Square GOF	98.31	5	< 0.001	Reject H ₀

Interpretation: since $p\text{-value} < 0.05$, this indicates that all barriers are not equally important. this indicates that certain sources, particularly industry, contribute significantly more compared to others.

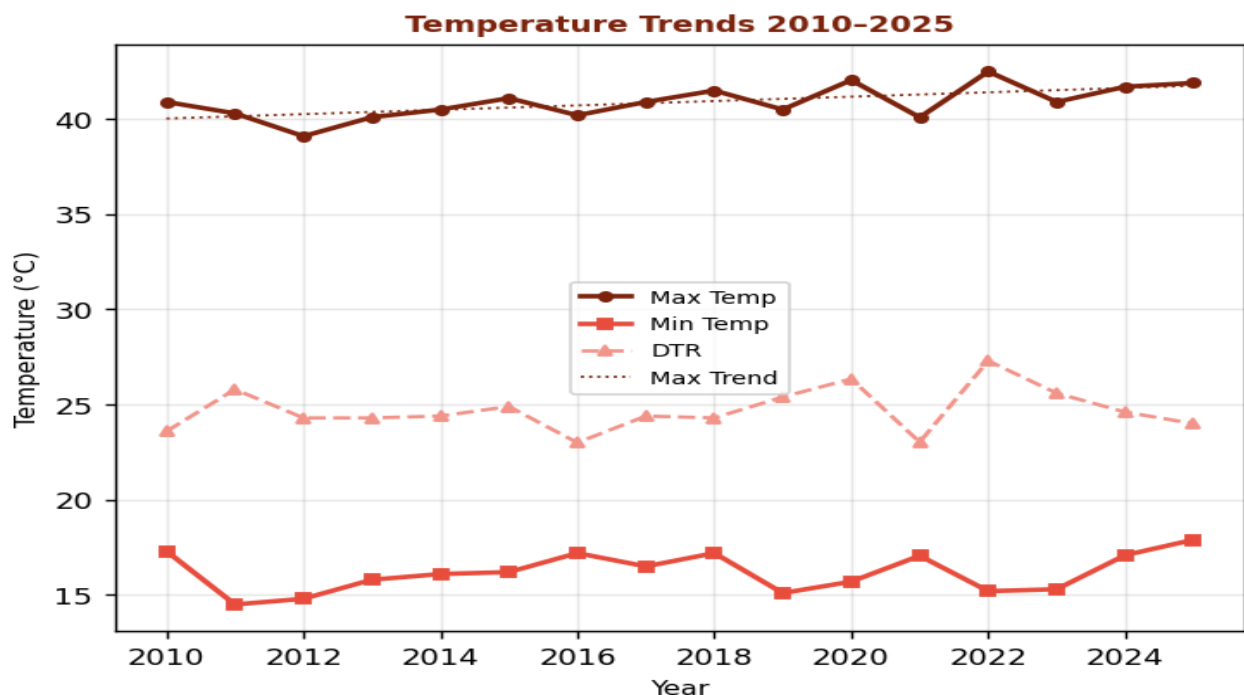
Frequency analysis identified the sources of carbon emission with the highest responses ,while chi square goodness of fit test confirmed that the observed differences among sources of carbon emission were statistically significant and not due to chance.

SECONDARY OBJECTIVE 1 – TEMPERATURE TREND ANALYSIS (2010– 2025)

Year	Max Temp (°C)	Min Temp (°C)	DTR (°C)
2010	40.9	17.3	23.6
2011	40.3	14.5	25.8
2012	39.1	14.8	24.3
2013	40.1	15.8	24.3
2014	40.5	16.1	24.4
2015	41.1	16.2	24.9
2016	40.2	17.2	23.0
2017	40.9	16.5	24.4
2018	41.5	17.2	24.3
2019	40.5	15.1	25.4
2020	42.05	15.7	26.35
2021	40.09	17.06	23.03
2022	42.5	15.2	27.3
2023	40.9	15.3	25.6
2024	41.7	17.09	24.61
2025	41.9	17.9	24.0

Both maximum and minimum temperatures show a rising trend from 2010 to 2025, with 2022 being the most extreme year (Max = 42.5°C). Rising night temperatures and increasing maximum temperatures together confirm progressive warming of the region, likely due to climate change

SECONDARY OBJECTIVE 1 – TEMPERATURE TREND ANALYSIS (2010–2025)



1. Maximum Temperature

Ranged from 39.1°C (2012) to 42.5°C (2022)

Started at 40.9°C (2010) and ended at 41.9°C (2025)

Shows a clear upward trend confirmed by the trend line in the graph

2022 was the hottest year recorded

Maximum temperatures are gradually but consistently rising over 16 years

2. Minimum Temperature

Ranged from 14.5°C (2011) to 17.9°C (2025)

Showed fluctuations but clearly rising toward 2024–2025

2025 recorded the highest minimum temperature (17.9°C)

Nights are getting warmer a strong sign of climate warming

3. DTR (Diurnal Temperature Range)

Ranged from 23.0°C (2016) to 27.3°C (2022)

No consistent trend — fluctuates year to year

DTR varies irregularly depending on yearly atmospheric conditions

SECONDARY OBJECTIVE 1 – SPEARMAN RANK CORRELATION TEST

Non-parametric Spearman Rank Correlation was applied to test whether a significant trend exists in each temperature series over time (2010–2025).

MAXIMUM TEMPERATURE:

H₀: no significant trend in annual maximum temperature .

H₁: significant trend in annual maximum temperature

MINIMUM TEMPERATURE:

H₀ : no significant trend in annual minimum temperature .

H₁: significant trend in annual minimum temperature .

DTR:

H₀: no significant trend in DTR .

H₁: significant trend in DTR .

Variable	Spearman ρ	p-value	Strength	Decision ($\alpha=0.05$)
Max Temperature	0.5653	0.0225	Moderate Positive	Reject H ₀
Min Temperature	0.2090	0.4373	Weak Positive	Fail to Reject H ₀
DTR	0.1993	0.4594	Weak Positive	Fail to Reject H ₀

Interpretation:

Maximum Temperature

Both the graph and Spearman test confirm that maximum temperature is significantly increasing This is clear evidence of Urban Heat Island Effect and Global Warming.

Minimum Temperature

The graph shows a slight rise, but according to the Spearman test, it is not statistically significant Therefore, we cannot confidently claim a trend for minimum temperature.

DTR

Neither the graph shows any clear trend, nor did the Spearman test prove one. DTR shows only random variations — there is no meaningful trend.

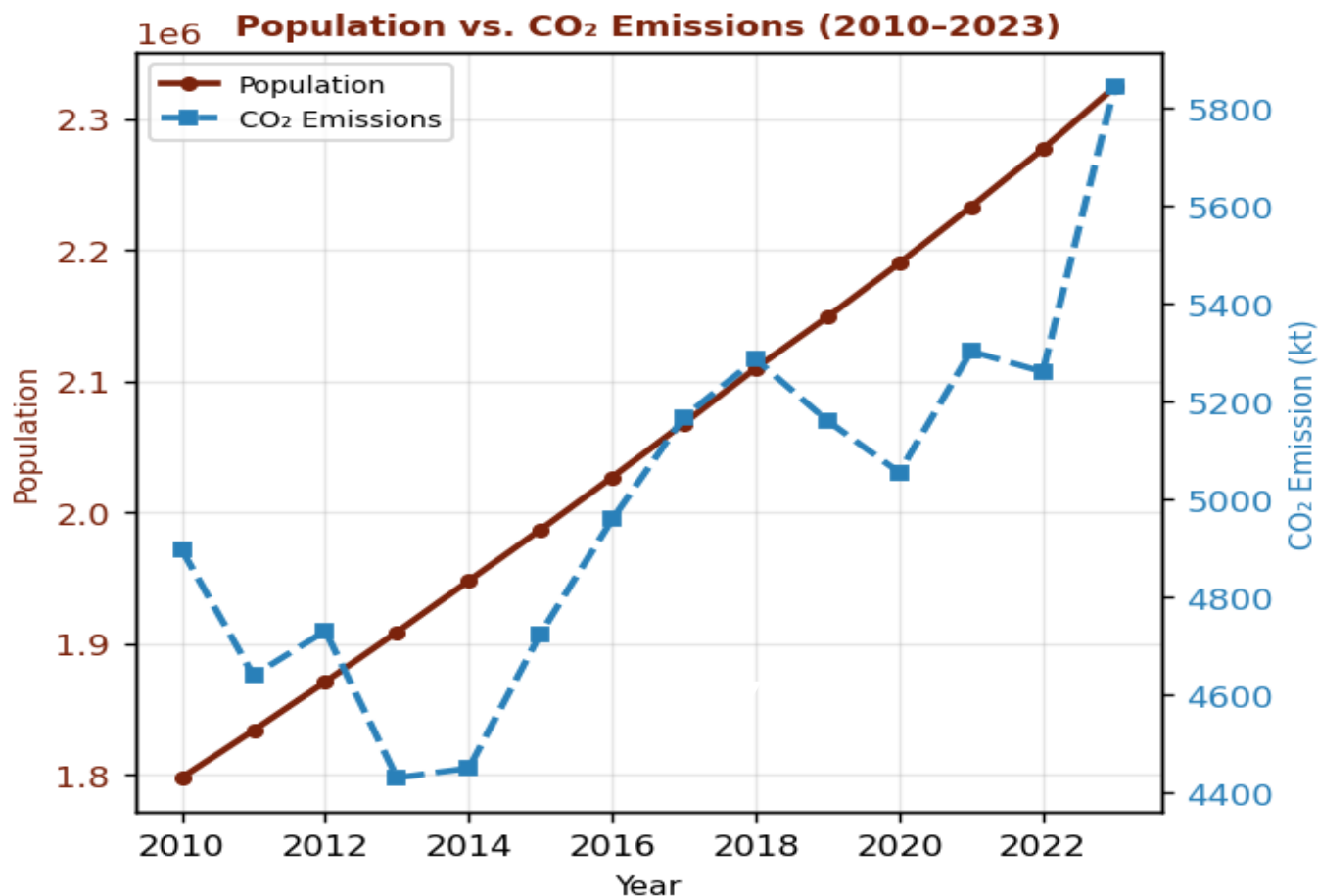
SECONDARY OBJECTIVE 2 – POPULATION GROWTH & CO₂ EMISSIONS

DATA ANALYSIS:

Year	Population	CO ₂ Emission (kt)
2010	1,798,000	4896.77
2011	1,834,000	4639.76
2012	1,871,000	4731.02
2013	1,909,000	4430.67
2014	1,948,000	4449.85
2015	1,987,000	4725.47
2016	2,027,000	4959.38
2017	2,068,000	5168.79
2018	2,110,000	5286.96
2019	2,149,000	5160.94
2020	2,190,000	5055.41
2021	2,233,000	5303.14
2022	2,277,000	5261.35
2023	2,324,000	5843.45

SECONDARY OBJECTIVE 2 – POPULATION GROWTH & CO₂ EMISSIONS

DATA ANALYSIS:



Interpretation:

As Vadodara's population grew from 1.79 million (2010) to 2.32 million (2023), CO₂ emissions overall increased from 4,897 kt to 5,843 kt.

However, Population growth and CO₂ emissions in Vadodara show a generally positive relationship over 2010–2023, but short-term deviations caused by policy changes, efficiency improvements, and global events like COVID-19 make the relationship non-linear and complex.

SECONDARY OBJECTIVE 2 –

CORRELATION ANALYSIS BETWEEN POPULATION AND CARBON EMISSION:

Year	Population	CO ₂ Emission (kt)
2010	1,798,000	4896.77
2011	1,834,000	4639.76
2012	1,871,000	4731.02
2013	1,909,000	4430.67
2014	1,948,000	4449.85
2015	1,987,000	4725.47
2016	2,027,000	4959.38
2017	2,068,000	5168.79
2018	2,110,000	5286.96
2019	2,149,000	5160.94
2020	2,190,000	5055.41
2021	2,233,000	5303.14
2022	2,277,000	5261.35
2023	2,324,000	5843.45

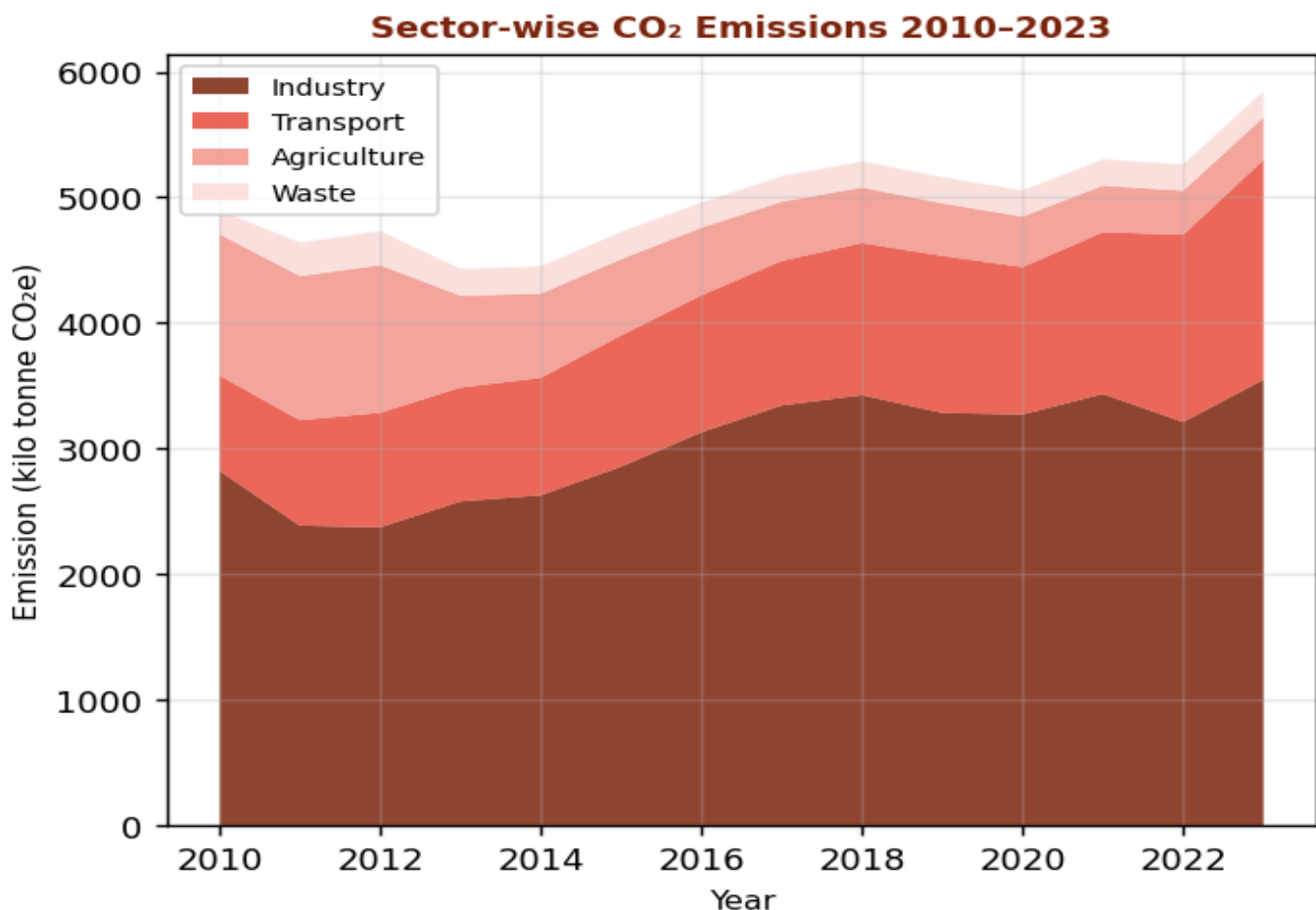
Correlation	Population	Total Emission
Population	1.000	0.824
Total Emission	0.824	1.000

Interpretation: The Pearson correlation coefficient $r = 0.824$ indicates a strong positive relationship between Population and CO₂ Emissions in Vadodara (2010–2023). This means as population increased from 1,798,000 to 2,324,000, CO₂ emissions also rose from 4,896 kt to 5,843 kt. The closer r is to 1, the stronger the relationship — and 0.824 confirms that population growth is a major driver of rising emissions.

SECONDARY OBJECTIVE 3 – SECTOR-WISE CARBON EMISSIONS (2010–2023)

Year	Industry	Transport	Agriculture	Waste
2010	2824	758	1125	189
2011	2387	840	1147	266
2012	2374	912	1174	271
2013	2580	907	731	213
2014	2630	935	670	215
2015	2857	1044	607	218
2016	3131	1090	538	200
2017	3344	1150	471	204
2018	3427	1211	441	208
2019	3285	1251	417	208
2020	3272	1173	402	208
2021	3436	1286	371	209
2022	3212	1491	350	208
2023	3550	1749	339	206

SECONDARY OBJECTIVE 3 – SECTOR-WISE CARBON EMISSIONS (2010–2023)

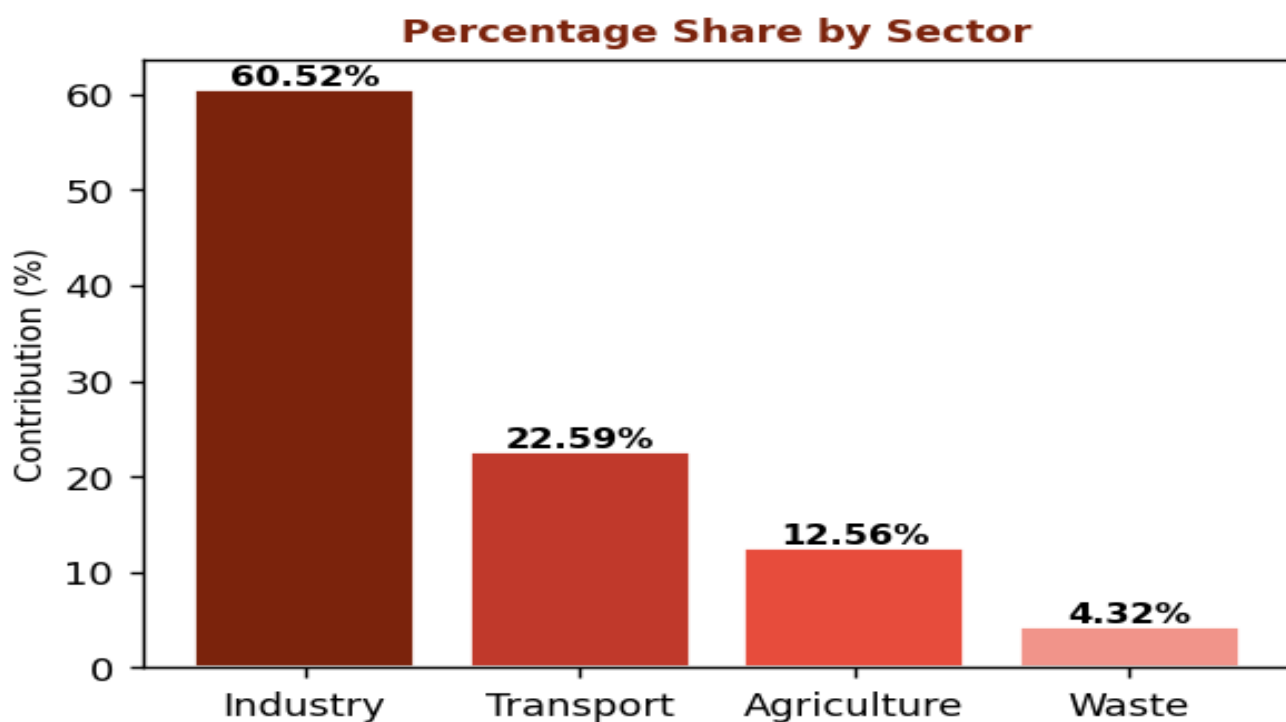


The stacked area chart represents the contribution of different sectors (Industry, Transport, Agriculture, and Waste) to total carbon emissions over time (2010–2023). It helps in visualizing both the total trend and the relative share of each sector

“Industry is the dominant contributor, transport is the fastest-growing sector, agriculture is declining, and waste has minimal impact on total emissions.”

SECONDARY OBJECTIVE 3 – SECTORAL CONTRIBUTION & INTERPRETATION

Sector	Total Emission (kt)	Contribution (%)
Industry	42,309.1	60.52%
Transport	15,796.0	22.59%
Agriculture	8,784.0	12.56%
Waste	3,023.8	4.32%
TOTAL	69,913.0	100%



Interpretation:

Overall sector wise carbon emission shows that industry and transport are the primary drives of Vadodara rising co2 emissions.

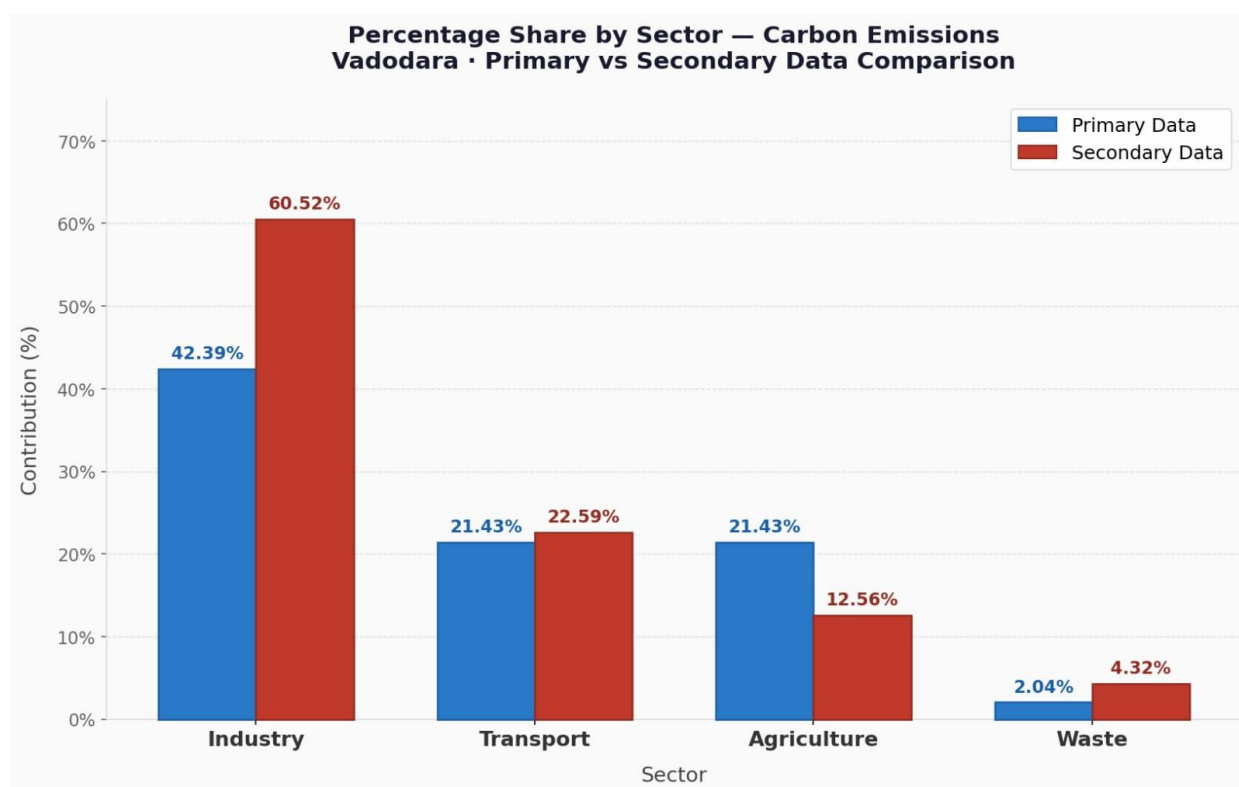
COMPARISON OF PRIMARY AND SECONDARY SECTOR WISE CARBON EMISSION :

Secondary wise sector emission

Sector	Contribution%
Industry	60.52
Transport	22.59
Agriculture	12.56
Waste	4.32

Primary wise sector emission

Sector	Contribution%
Industry	42.39
Transport	21.43
Agriculture	21.43
Waste	2.04



Interpretation:

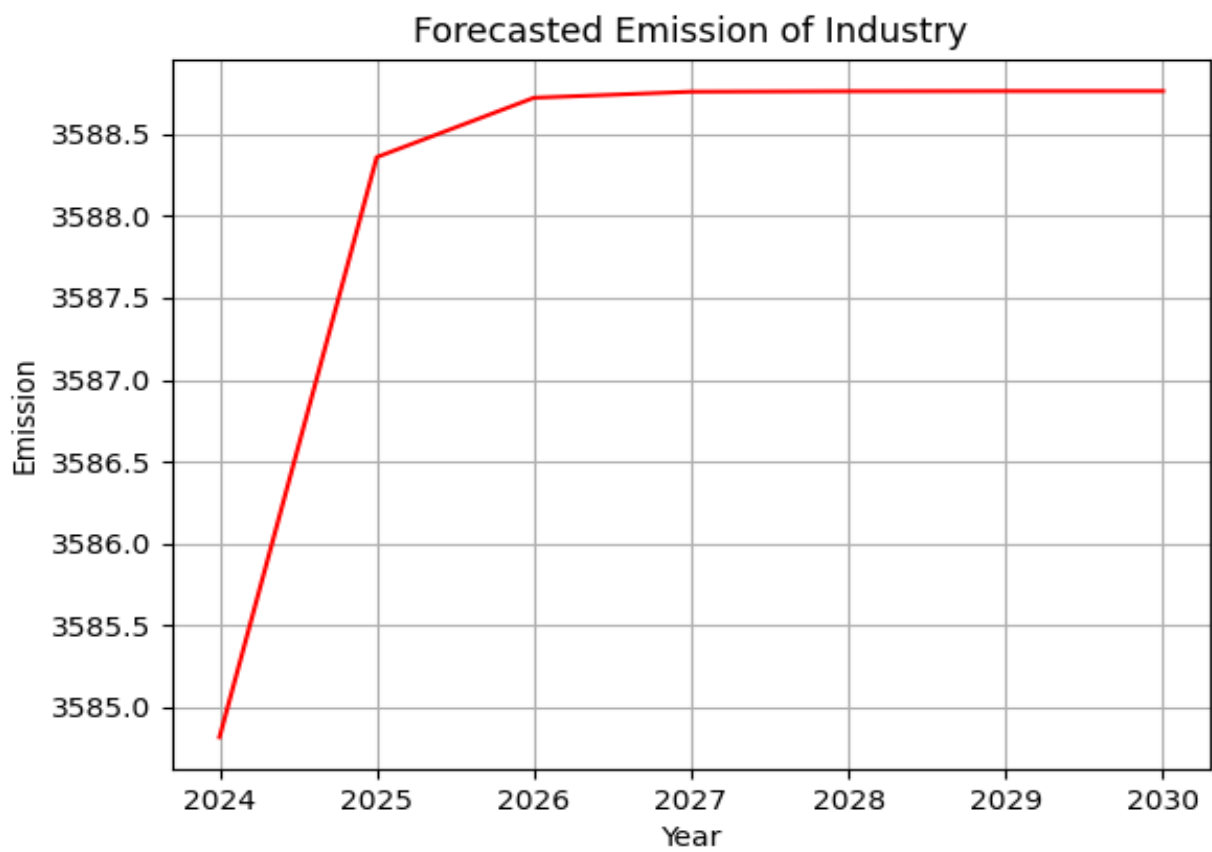
Industry is the biggest emitter in both datasets (Primary: 42.39%, Secondary: 60.52%), confirming it as the main driver of CO₂ emissions. Transport shows the closest match between both data sources (~21–23%), proving consistency. Agriculture shows the biggest difference — primary data recorded much higher (21.43%) than secondary (12.56%), suggesting ground reality differs from published data. Waste is the least contributor in both.

SECONDARY OBJECTIVE 3 – EMISSION FORECASTS (2024–2027)

Forecasting was performed using simple linear trend models fitted to 2010–2023 data. The projections provide a reference scenario assuming continuation of historical trends.

FORECASTED EMISSION OF INDUSTRY:

Year	INDUSTRY
2024	3584.816
2025	3588.360
2026	3588.723
2027	3588.760
2028	3588.763
2029	3588.764
2030	3588.764

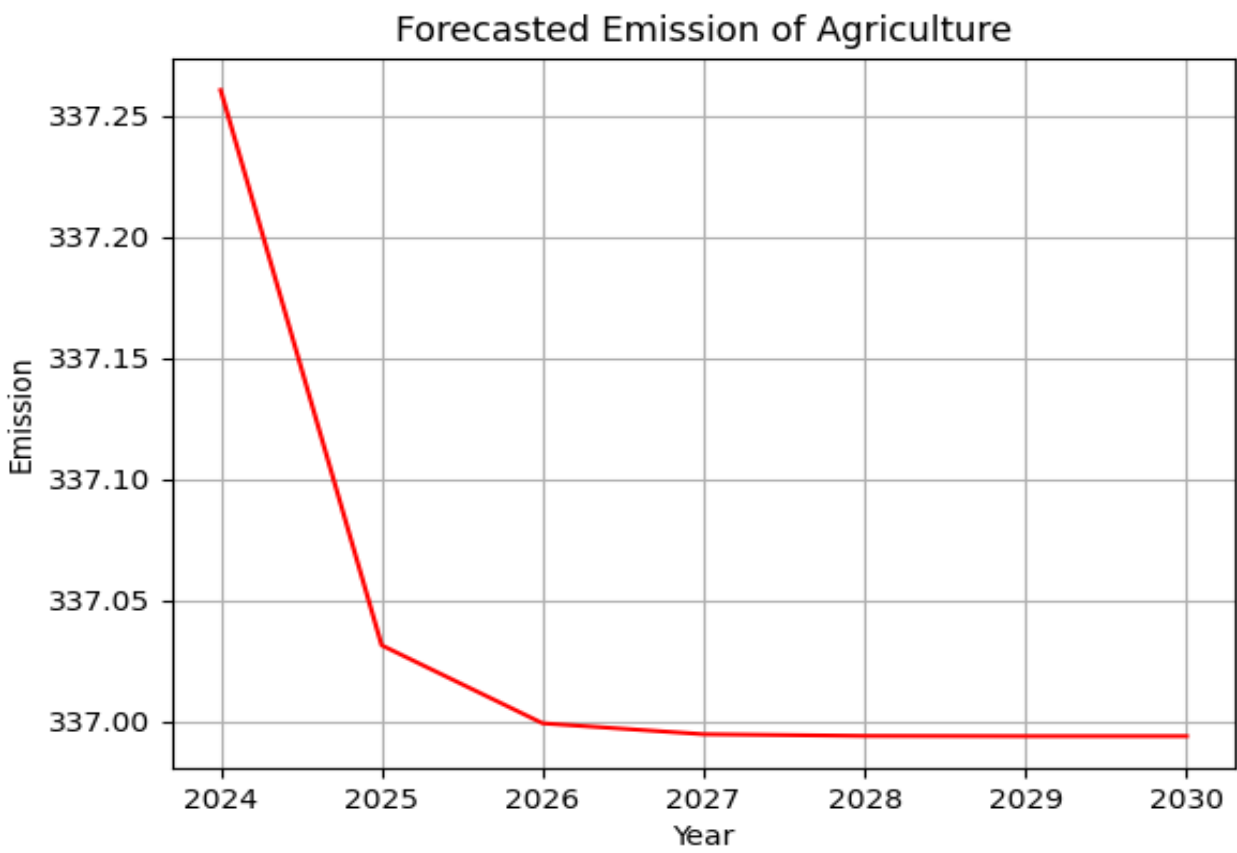


SECONDARY OBJECTIVE 3 –

Forecasting was performed using simple linear trend models fitted to 2010–2023 data. The projections provide a reference scenario assuming continuation of historical trends.

FORECASTED EMISSION OF AGRICULTURE:

Year	AGRICULTURE
2024	337.2605
2025	337.0314
2026	336.9992
2027	336.9947
2028	336.9940
2029	336.9939
2030	336.9939

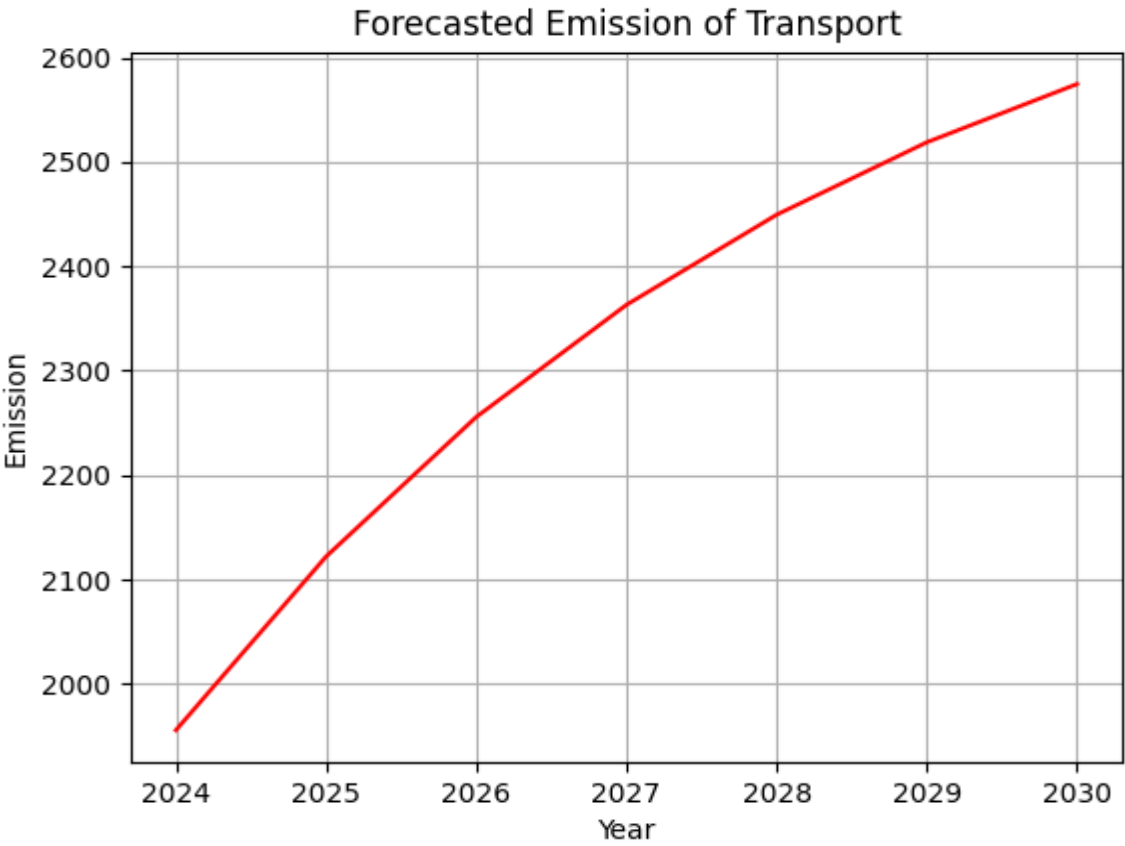


SECONDARY OBJECTIVE 3 –

Forecasting was performed using simple linear trend models fitted to 2010–2023 data. The projections provide a reference scenario assuming continuation of historical trends.

FORECASTED EMISSION OF TRANSPORT:

Year	TRANSPORT
2024	1955.628
2025	2121.927
2026	2255.573
2027	2362.977
2028	2449.293
2029	2518.661
2030	2574.409

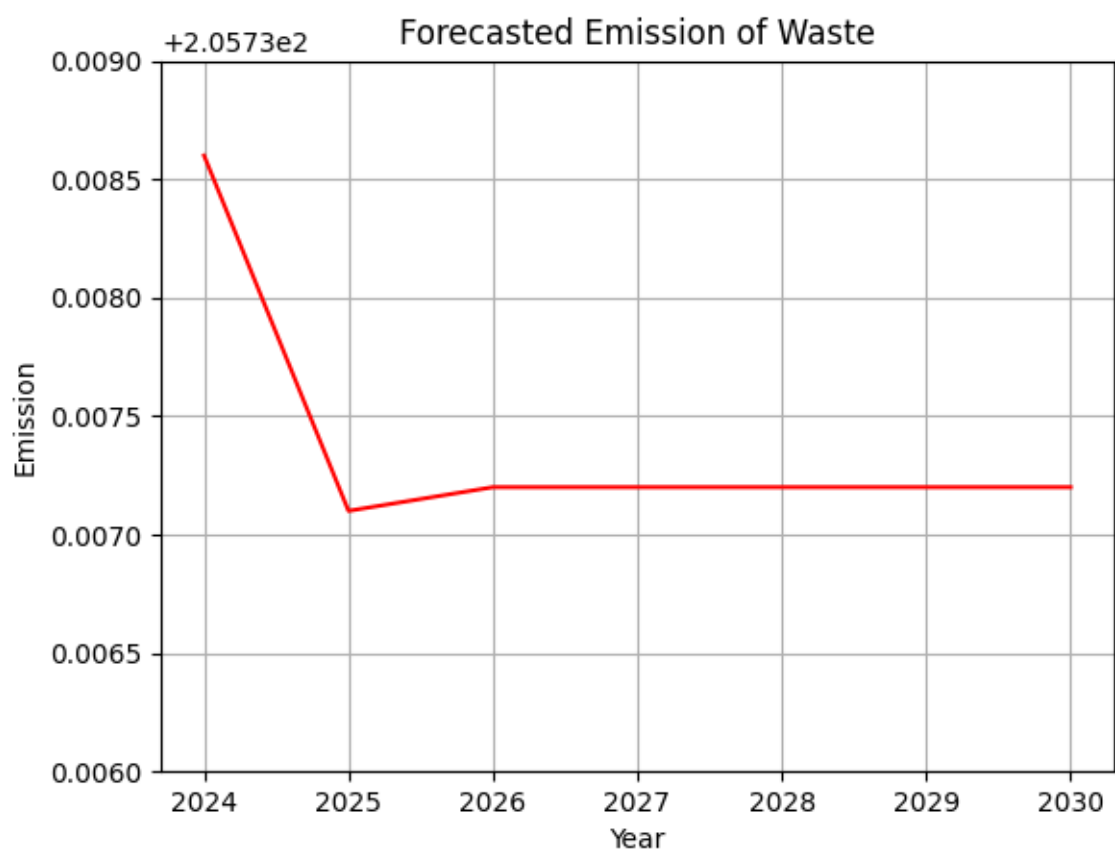


SECONDARY OBJECTIVE 3 –

Forecasting was performed using simple linear trend models fitted to 2010–2023 data. The projections provide a reference scenario assuming continuation of historical trends.

FORECASTED EMISSION OF WASTE:

Year	WASTE
2024	205.7386
2025	205.7371
2026	205.7372
2027	205.7372
2028	205.7372
2029	205.7372
2030	205.7372



CONCLUSION

- 1) **PO1 – Education & Awareness:** Education level significantly influences climate change awareness . Post-graduates show high awareness while school-educated respondents are predominantly in the low awareness category. Graduate respondents show no significant bias.
- 2) **PO2 – Awareness & Eco-behaviour:** A strong positive association exists between awareness level and recycling behaviour . Higher awareness leads to more frequent eco-friendly behaviour.
- 3) **PO3 – Barriers:** The primary barriers are high cost (54.6%), lack of infrastructure (52.6%), and limited awareness (37.5%). Addressing these through policy interventions is essential.
- 4) **PO4 – Emission Sources:** Industry (38.3%) is the most commonly identified emission source, but Agriculture and Waste are significantly underestimated .
- 5) **SO1 – Temperature Trend:** Maximum temperature in Vadodara shows a statistically significant upward trend from 2010–2025. Minimum temperature and DTR show no significant trend.
- 6) **SO2 – Population & Emissions:** Strong positive correlation between population growth and CO₂ emissions. Population growth is identified as a major driver of Vadodara's rising emissions.
- 7) **SO3 – Sector Emissions:** Industry accounts of total emissions. Transport is the fastest-growing sector. Agriculture contribution is declining. Emission forecasts project continued rise without intervention.

Python code:

```
import numpy as np
import pandas as pd
from scipy.stats import chi2_contingency
# chi square test of independence(education vs awareness)---primary objective 1
# Observed table
data = np.array([
    [22, 55, 41], # Graduate
    [13, 24, 38], # Post Graduate
    [21, 26, 8]   # School
])
df = pd.DataFrame(data,
                  index=["Graduate", "Post Graduate", "School"],
                  columns=["Low", "Moderate", "High"])
chi2, p, dof, expected = chi2_contingency(df)
print("Chi-square:", chi2)
print("p-value:", p)
print("Degrees of freedom:", dof)
print("Expected frequencies:\n", expected)
```

```
Chi-square: 22.179710897064492
p-value: 0.0001845687121460064
Degrees of freedom: 4
Expected frequencies:
[[26.64516129 49.95967742 41.39516129]
 [16.93548387 31.75403226 26.31048387]
 [12.41935484 23.28629032 19.29435484]]
```

goodman krushkal gamma(awareness vs recycling behaviour)---primary objective 2

```
import numpy as np
# Awareness vs Recycling table
T = np.array([
    [25,21,12,4,1],
    [5,10,15,9,6],
    [2,6,15,25,34]
])
C = D = 0
for i in range(len(T)):
    for j in range(len(T[0])):
        for k in range(i+1, len(T)):
            for l in range(j+1, len(T[0])):
                C += T[i,j] * T[k,l]
                D += T[k,j] * T[i,l]
gamma = (C - D) / (C + D)
print("Gamma =", gamma)
```

```
Gamma = 0.7264112903225807
```

```

# goodman krushkal gamma(awareness vs saving electricity behaviour)---primary objective 2
import numpy as np
# Awareness vs saving electricity
T = np.array([
    [13,17,16,10,7],
    [2,5,12,14,12],
    [1,4,11,25,41]
])
C = D = 0
for i in range(len(T)):
    for j in range(len(T[0])):
        for k in range(i+1, len(T)):
            for l in range(j+1, len(T[0])):
                C += T[i,j] * T[k,l]
                D += T[k,j] * T[i,l]
gamma = (C - D) / (C + D)
print("Gamma =", gamma)

```

Gamma = 0.5963302752293578

```

#goodman keushkal gamma (awareness vs public transport)--primary objective 2
import numpy as np
# Awareness vs saving electricity
T = np.array([
    [19,19,13,8,4],
    [6,9,13,10,7],
    [6,10,19,22,25]
])
C = D = 0
for i in range(len(T)):
    for j in range(len(T[0])):
        for k in range(i+1, len(T)):
            for l in range(j+1, len(T[0])):
                C += T[i,j] * T[k,l]
                D += T[k,j] * T[i,l]
gamma = (C - D) / (C + D)
print("Gamma =", gamma)

```

Gamma = 0.47010698552548774

```
#chi square goodness of fit(barriers)--- primary objective 3
from scipy.stats import chisquare
observed = np.array([166,160,114,91,14])
expected = np.repeat(sum(observed)/len(observed), len(observed))
chi2, p = chisquare(observed, expected)
print("Chi-square:", chi2)
print("p-value:", p)
```

Chi-square: 139.6697247706422
p-value: 3.321633226252264e-29

```
#spearman rank correlaion(maximum temperature trend)==secondary objective 1
from scipy.stats import spearmanr
years = list(range(2010, 2026))
max_temp = [40.9,40.3,39.1,40.1,40.5,41.1,40.2,40.9,41.5,40.5,42.05,40.09,42.5,40.9,41.7,41.9]
rho, p = spearmanr(years, max_temp)
print("Spearman rho:", rho)
print("p-value:", p)
```

Spearman rho: 0.5653175019364274
p-value: 0.022484676071615476

```
#spearman rank correlaion(minimum temperature trend)==secondary objective 1
from scipy.stats import spearmanr
years = list(range(2010, 2026))
max_temp = [17.3,14.5,14.8,15.8,16.1,16.2,17.2,16.5,17.2,15.1,15.7,17.06,15.2,15.3,17.09,17.9]
rho, p = spearmanr(years, max_temp)
print("Spearman rho:", rho)
print("p-value:", p)
```

Spearman rho: 0.20897724568536738
p-value: 0.43731360129472274

```
#spearman rank correlaion(dtr)==secondary objective 1
from scipy.stats import spearmanr
years = list(range(2010, 2026))
max_temp = [23.6,25.8,24.3,24.3,24.4,24.9,23.0,24.4,24.3,25.4,26.35,23.03,27.3,25.6,24.61,24.0]
rho, p = spearmanr(years, max_temp)
print("Spearman rho:", rho)
print("p-value:", p)
```

Spearman rho: 0.19926334924652142
p-value: 0.4593804641854442

```
#pearson correlation cofficient(population vs co2)==secondary objective 3
from scipy.stats import pearsonr
population = [1798000,1834000,1871000,1909000,1948000,1987000,2027000,
              2068000,2110000,2149000,2190000,2233000,2277000,2324000]
co2 = [4896.77,4639.76,4731.02,4430.67,4449.85,4725.47,4959.38,
        5168.79,5286.96,5160.94,5055.41,5303.14,5261.35,5843.45]
r, p = pearsonr(population, co2)
print("Pearson r:", r)
print("p-value:", p)
```

Pearson r: 0.8244790880449441
p-value: 0.0002859283052584703

```
# chi square goodness of fit(emission resources)---primary objective 4
observed = np.array([75,42,39,14,12,7,4,3])
expected = np.repeat(sum(observed)/len(observed), len(observed))
chi2, p = chisquare(observed, expected)
print("Chi-square:", chi2)
print("p-value:", p)
```

Chi-square: 184.57142857142856
p-value: 2.1080410720472095e-36

R-CODE:

```
industry <- c(
  2824.16, 2386.748, 2373.869, 2580.038,
  2629.724, 2857.186, 3131.25, 3344.418,
  3426.676, 3284.779, 3272.058, 3436.351,
  3211.686, 3550.181)
industry

industry_ts <- ts(industry, start = 2010, frequency = 1)
industry_ts

# ARIMA with differencing
ar_model <- arima(industry_ts, order = c(1, 1, 0))
summary(ar_model)

# Forecast 2024-2030
forecast_values <- predict(ar_model, n.ahead = 7)

data.frame(
  Year = 2024:2030,
  Forecast = as.numeric(forecast_values$pred))
```

Year <int>	Forecast <dbl>
2024	3584.816
2025	3588.360
2026	3588.723
2027	3588.760
2028	3588.763
2029	3588.764
2030	3588.764

```

transport <- c(
  758.0595, 839.5434, 912.2448, 906.545,
  934.6911, 1043.94, 1089.624, 1150.022,
  1211.227, 1250.533, 1173.339, 1286.342,
  1491.213, 1748.699)
transport

transport_ts <- ts(transport, start = 2010, frequency = 1)
transport_ts

# ARIMA with differencing
ar_model <- arima(transport_ts, order = c(1, 1, 0))
summary(ar_model)

# Forecast 2024-2030
forecast_values <- predict(ar_model, n.ahead = 7)

data.frame(
  Year = 2024:2030,
  Forecast = as.numeric(forecast_values$pred))

```

Year <int>	Forecast <dbl>
2024	1955.628
2025	2121.927
2026	2255.573
2027	2362.977
2028	2449.293
2029	2518.661
2030	2574.409

```

agriculture <- c(
  1125.27, 1147.35, 1174.11, 730.82,
  670.03, 606.6, 538.32, 470.55,
  441.46, 417.23, 401.65, 371.27,
  350.48, 338.89)
agriculture

agriculture_ts <- ts(agriculture, start = 2010, frequency = 1)
agriculture_ts

# ARIMA with differencing
ar_model <- arima(agriculture_ts, order = c(1, 1, 0))
summary(ar_model)

# Forecast 2024-2030
forecast_values <- predict(ar_model, n.ahead = 7)

data.frame(
  Year = 2024:2030,
  Forecast = as.numeric(forecast_values$pred))

```

Year <int>	Forecast <dbl>
2024	337.2605
2025	337.0314
2026	336.9992
2027	336.9947
2028	336.9940
2029	336.9939
2030	336.9939

```

waste <- c(
  189.28, 266.12, 270.8, 213.27,
  215.4, 217.74, 200.19, 203.8,
  207.6, 208.4, 208.36, 209.18,
  207.97, 205.68)
waste

waste_ts <- ts(waste, start = 2010, frequency = 1)
waste_ts

# ARIMA with differencing
ar_model <- arima(waste_ts, order = c(1, 1, 0))
summary(ar_model)

# Forecast 2024-2030
forecast_values <- predict(ar_model, n.ahead = 7)

data.frame(
  Year = 2024:2030,
  Forecast = as.numeric(forecast_values$pred))

```

Year <int>	Forecast <dbl>
2024	205.7386
2025	205.7371
2026	205.7372
2027	205.7372
2028	205.7372
2029	205.7372
2030	205.7372

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THANK YOU

*Public Awareness and Behavioral Intentions for Climate Change
Mitigation of Vadodara City*

Mahak Garg | Jaymeet Modi | Darshan Bokadiya | Rutvik Chaudhari

Guided by: Dr. Manisha Pandya

Department of Statistics | M.S. University of Baroda | 2025–26